

UI Design Decisions and User Flow

Design Decisions

The user interface (UI) was designed with simplicity and clarity as the main goals, ensuring that both technical and non-technical users can interact with the system without prior training. The key design decisions were:

1.Minimalist Layout

- A clean and uncluttered interface to avoid distractions.
- Only essential elements such as the title, description, text input, and prediction outputs are displayed.

2.Streamlit Framework

- Chosen for its ease of development and ability to quickly create interactive web apps using Python.
- Allows seamless integration of machine learning models without requiring front-end development skills.

3.Two-Model Comparison

- The UI displays predictions from both Model_A (trained on balanced data) and Model_B (trained on imbalanced data).
- This allows the user to compare model performance on the same review instantly.

4.Interactive Input Area

- A multi-line text box is provided for entering customer reviews.
- This allows flexibility for short or long review inputs.

5.Clear Call-to-Action

- A single "Predict Ratings" button triggers the prediction process.
- Avoids confusion by limiting unnecessary actions.

User Flow

The interaction between the user and the system follows a simple, linear flow:

1.App Launch

- The user opens the Streamlit web application.
- The application title and brief description are displayed to inform the purpose of the system.

2.Input Review

-The user enters a product review into the text area provided.

3.Prediction Trigger

-The user clicks on the “Predict Ratings” button.

4.Model Processing

-The entered text is passed to both models for prediction.

-The models output the predicted rating for the review.

5.Results Display

-Predictions from both models are displayed clearly:

Model_A: Rating from the balanced-trained logistic regression model.

Model_B: Rating from the imbalanced-trained model.

6.User Interpretation

-The user compares the results and can enter another review for testing if desired.